

Birla Institute of Technology and Science, Pilani
First Semester, 2015-16
CHE F212, Fluid Mechanics

Quiz 6(open Book)

30/09/15

15 min, 5 points

NAME:

ID:

Section:

The pressure drop between two taps separated in the streamwise direction by 9 m in a horizontal, fully developed channel flow of water is 6.9 kPa. The cross section of the channel is a 25 mm × 240 mm rectangle. Calculate the average wall shear stress.

Solution:

Basic equation $F_x = F_{S_x} + F_{B_x} = \frac{\partial}{\partial t} \int_{CV} u \rho dV + \int_{CS} u \rho \vec{V} \cdot d\vec{A}$ (Eq. 4.18a)

Assumptions: 1) Horizontal pipe 2) Steady flow 3) Fully developed flow

With these assumptions, the x momentum equation becomes

$$p_1 \cdot W \cdot H + \tau_w \cdot 2 \cdot L \cdot (W + H) - p_2 \cdot W \cdot H = 0 \quad \text{or} \quad \tau_w = (p_2 - p_1) \cdot \frac{W \cdot H}{2 \cdot (W + H) \cdot L} \quad \tau_w = -\Delta p \cdot \frac{\frac{H}{L}}{2 \cdot \left(1 + \frac{H}{W}\right)}$$

$$\tau_w = -\frac{1}{2} \times 6.9 \times 10^3 \frac{\text{N}}{\text{m}^2} \times \frac{25 \text{ mm} \times \frac{1 \text{ m}}{1000 \text{ mm}}}{9 \text{ m}} \times \frac{1}{1 + \frac{240 \text{ mm} \times \frac{1 \text{ m}}{1000 \text{ mm}}}{9 \text{ m}}} \quad \tau_w = -9.33 \text{ Pa}$$

Since $\tau_w < 0$, it acts to the left on the fluid, to the right on the channel wall.